

SIH 1505 — INDUSTRIAL EMERGENCY PRE-PLAN

INCIDENT ID: INC-SUB-01-26K

DOCUMENT VERSION: v1.0 [AUTHORIZED
(PROTOTYPE DEMO)]

GENERATED: 2026-08-15 17:27:11 UTC

PREPARED BY: SIH-1505 Decision Support Engine

FACILITY: PetroChem Complex Alpha - Unit 04

PROTOTYPE FACILITY REF: PCH-ALPHA-04 (Demo Facility — Non-Statutory Evaluation)

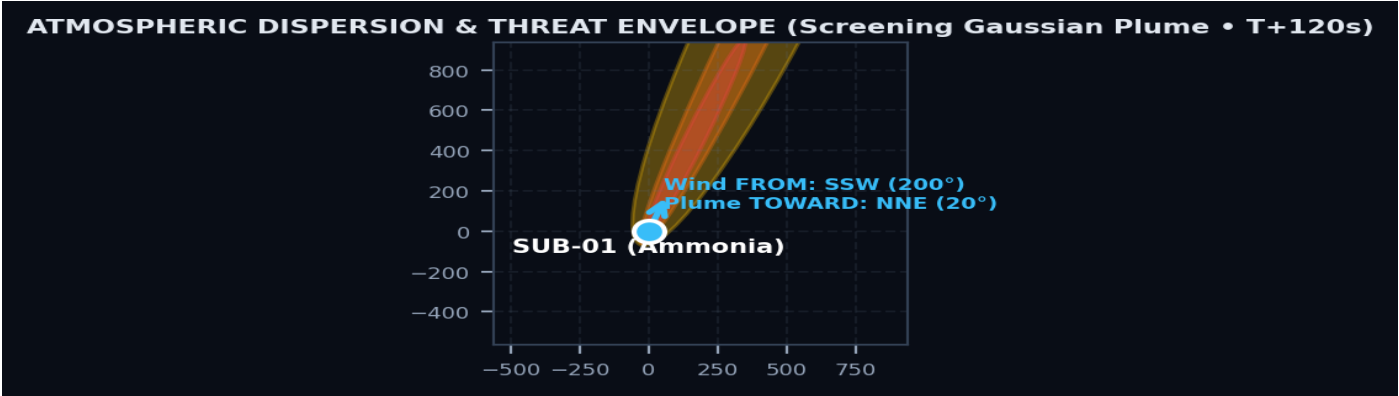
INCIDENT SEVERITY: MODERATE (SCORE: 41.1/100) • GOVERNANCE: AUTHORIZED BY DUTY HSE OFFICER (Chief Safety Officer (HSE))

1. INCIDENT SOURCE, CHEMICAL SDS & METEOROLOGICAL PROFILE

Incident Type:	Pipeline Leak	Source Asset:	SUB-01 (21.6885, 72.5705)
Hazardous Substance:	Ammonia (Anhydrous)	Release Emission Rate:	15.0 kg/s (Duration: 30 min)
Meteorological Wind Vector:	26.5 km/h • FROM: SSW (200°)	Plume Propagation Vector:	Downwind Travel • TOWARD: NNE (20°)
Ambient Temperature:	28.5°C	Weather Telemetry Source:	DEMO (Scenario Preset) • Stability: Class D

2. HAZARD THREAT ZONES (SCREENING GAUSSIAN DISPERSION ENVELOPE)

Zone Tier	Threshold Criterion	Downwind Reach	Crosswind Width	Threat Area
Red Threat Zone	ERPG-3 / High Lethality (750.0 ppm)	1101.8 m	75.4 m	54,784 m²
Orange Threat Zone	ERPG-2 / Irreversible Injury (150.0 ppm)	2743.9 m	170.1 m	305,355 m²
Yellow Threat Zone	ERPG-1 / Mild Irritation (25.0 ppm)	3000.0 m	390.1 m	677,252 m²



3. POPULATION & INFRASTRUCTURE EXPOSURE IMPACT

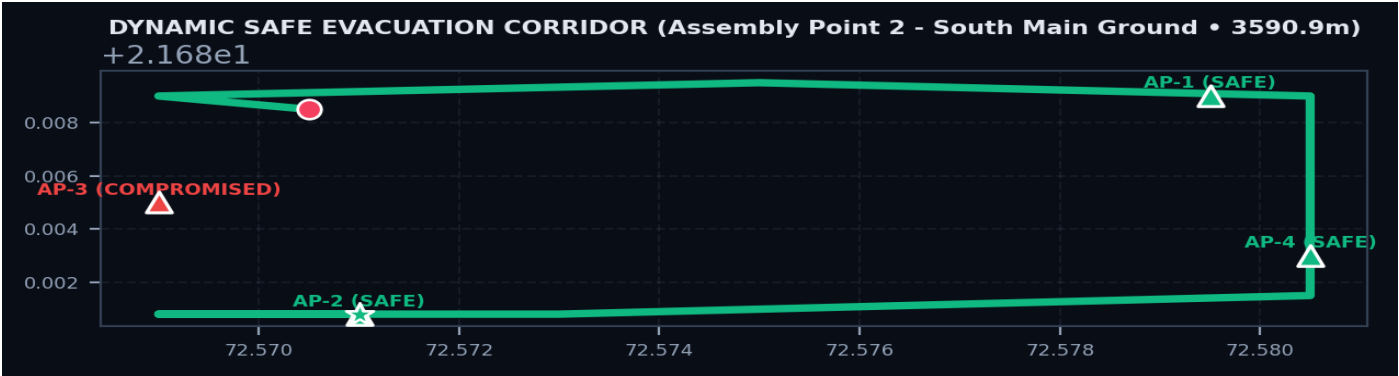
Total Workers on Site:	28 Personnel Exposure Assessment: No active seeded worker coordinates intersected the calculated threat envelopes at simulation time.	Lethal Zone (Red) Exposed:	0 Workers
Severe Zone (Orange) Exposed:	0 Workers	Caution Zone (Yellow) Exposed:	0 Workers
Compromised Plant Assets:	1 of 16 Units	Severed Road Segments:	3 of 10 Segments

4. DYNAMIC SAFE EVACUATION CORRIDOR & MUSTER DIRECTIVES

Designated Safe Assembly Point:	Assembly Point 2 - South Main Ground	Exit Perimeter Gate:	Gate 2 - South Commercial & Tanker Logistics Gate
Total Safe Egress Distance:	3590.9 meters	Estimated Walk Time:	49.9 minutes (at 1.2 m/s) ■ LONG EGRESS — HUMAN REVIEW REQUIRED (Threshold: ≥ 30m)
Multi-Factor Scoring:	Safety: 95% Dist: 10% Rank: 0.610	Route Clearance Status:	CLEAR
Selection Rationale:	Selected as optimal Upwind/Crosswind egress corridor (156° clearance from plume axis) providing shortest uncompromised route to Assembly Point 2 - South Main Ground (Safety Score: 0.95, Composite Score: 0.61).	Avoided Blocked Roads:	Perimeter Ring Road - West, Chemical Sector Cross Road (East-West North), Process Area Cross Road (East-West South)

Evaluated Muster Point Alternatives & Rejection Analysis:

Candidate Assembly Point	Exit Gate	Distance	Safety	Status / Rejection Reason
ROUTE-AP-2 (AP-2)	GATE-2	3590.9 m	95%	SELECTED: None
ROUTE-AP-1 (AP-1)	GATE-1	1461.4 m	85%	VIABLE_BACKUP: Lower safety/composite score (0.55 vs 0.61)
ROUTE-AP-4 (AP-4)	GATE-3	2020.0 m	70%	REJECTED: Internal access roads severed; requires outer peripheral detour
ROUTE-AP-3 (AP-3)	GATE-2	609.4 m	5%	REJECTED: Assembly Point inside toxic hazard envelope (Red Lethal Zone)



5. TACTICAL EMERGENCY RESOURCE DEPLOYMENT & SUPPRESSION STRATEGY

DECISION-SUPPORT RECOMMENDATION — REQUIRES SITE/HSE VALIDATION: All tactical quantities, PPE ensembles, suppression demands, and response actions are prototype computational recommendations and must be validated against facility ERDMP and competent safety authorities before operational deployment.

Upwind Staging Standoff:	330.5 meters	Isolation Cordon Radius:	1377.2 meters
Firewater Demand:	6,100 LPM	Foam Concentrate Demand:	0 Liters (AFFF 3%)
Mandatory Entry PPE:	Level A Fully Encapsulated Gas-Tight Suit (Trellechm) with positive-pressure SCBA (NIOSH Certified)	Calculation Basis:	Calculated from release rate 15.0 kg/s, 1 vulnerable assets, and Ammonia (Anhydrous) SDS toxicity/flammability

Unit Name & Type	Station	Staging Geolocation	Transit	ETA	Priority
High-Volume Water Curtain Bowser High-Volume Vapor Cloud Knockdown & Gas Absorption	FS-01	Crosswind Knockdown Sector (110° Vector)	682.5 m	2.4m	IMMEDIATE
Hazmat Rapid Response Unit (Level A/B Ensembles) Hot Zone Isolation & Emergency Valve Clamping	FS-01	Incident Hot-Zone Control Post (20° Upwind)	1022.9 m	3.3m	IMMEDIATE
Fire Tender Alpha (10,000L Multi-Purpose Water/Foam) Exposure Protection & Boundary Water Curtain Backup	FS-01	Upwind Incident Staging Post (Bearing 20°, 330m Standoff)	1033.6 m	3.3m	HIGH
Industrial Emergency Response Team (ERT Alpha) Cordon Enforcement, Road Junction Barricading & Headcount Control	CR-01	Main Outer Access Gate & Security Command	303.6 m	1.3m	SUPPORT

Advanced Cardiac & Toxic Resuscitation Ambulance Precautionary Medical Standby at Muster Point	MED-01	Medical Triage at Assembly Point 2 - South Main Ground	465.5 m	1.8m	STANDBY
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6. INCIDENT COMMAND EXECUTIVE SUMMARY SNAPSHOT

Risk:	41.1/100 — MODERATE	Chemical:	Ammonia (Anhydrous)
Personnel Exposed:	0 workers (Red: 0)	Compromised Assets:	1 of 16 units
Blocked Roads:	3 segments	Primary Muster:	Assembly Point 2 - South Main Ground
Exit Gate:	Gate 2 - South Commercial & Tanker Logistics Gate	Egress Metrics:	3590.9 m / 49.9 min (■ LONG EGRESS)
Lead Tactical Resource:	High-Volume Water Curtain Bowser (IMMEDIATE, ETA: 2.4m)	Firewater Demand:	6,100 LPM Water (0 L AFFF Foam)

7. STANDARD OPERATING PROCEDURE (SOP) PHASED ACTION DIRECTIVES

Phase 1: Initial Alarm, Isolation & Personnel Evacuation (0 - 5 Minutes)

- Sound Plant Emergency Siren (Continuous Alarm for Ammonia (Anhydrous) Pipeline Leak).
- Actuate Emergency Shutdown (ESD) for SUB-01 and isolate upstream pipeline valves.
- Enforce mandatory crosswind evacuation of all Sector personnel towards Assembly Point 2 - South Main Ground.
- ERT Team establishes 1377.2m initial cordon perimeter around SUB-01.

Phase 2: Source Mitigation & Tactical Exposure Protection (5 - 15 Minutes)

- Position Fire Tender & Bowser at Upwind Staging Post (330m standoff, bearing 20°).
- Establish 6,100 LPM water curtain coverage to suppress Ammonia (Anhydrous) vapors and cool 1 adjacent assets.
- Deploy Hazmat Entry Squad in Level A Fully Encapsulated Gas-Tight Suit for hot-zone valve isolation and clamp seal.
- Verify automated fixed water deluge systems on neighboring storage vessels.

Phase 3: Containment, Medical Triage & Incident De-escalation (15 - 30 Minutes)

- Complete headcount verification at Assembly Point 2 - South Main Ground (Reconcile against 0 exposed workers).
- Ambulance team initiates emergency triage and decontamination for exposed operators.
- Contain all contaminated firefighting water in industrial effluent retention sump; prevent municipal drain escape.
- HSE Lead Commander compiles situation briefing for District Disaster Management Authority (DDMA).

8. DATA TRACEABILITY & SCREENING MODEL ASSUMPTIONS

Dispersion Physics:	Screening Gaussian Approximation with Pasquill-Gifford stability parameters	Human Governance:	Mandatory HSE human review and authorization required prior to operational execution
Weather Telemetry:	DEMO Telemetry (Scenario Preset) • Recorded at 28.5°C, 26.5 km/h	Statutory Scope:	Decision-support prototype only; non-certified under statutory OISD/ALOHA rules
Terrain Modeling:	Simplified 2D/3D flat terrain representation with standard surface roughness (z ₀ = 0.5m)	Road & Worker Graph:	Prototype facility CAD/GIS network with obstacle avoidance pathfinding

9. REGULATORY DISCLAIMER & PROTOTYPE NOTICE

PROTOTYPE DECISION SUPPORT — NON-CERTIFIED COMPUTATIONAL WORKFLOW:

This emergency pre-plan document is prepared automatically by the SIH 1505 Decision Support Engine. Dispersion envelopes, routing corridors, and tactical dispatches are computed for hackathon demonstration and decision-support prototyping only. This software does NOT claim certified ALOHA equivalence or statutory compliance. Operational tactical actions require validation against the facility's approved Emergency Response and Disaster Management Plan (ERDMP), applicable OISD/PESO standards, and competent statutory safety authorities.

10. HUMAN-IN-THE-LOOP PRE-PLAN AUTHORIZATION & ENDORSEMENT

DOCUMENT STATUS: AUTHORIZED (PROTOTYPE DEMO)

Document Version: v1.0

Prepared by: SIH-1505 Decision Support Engine

Demonstration Approver: **Duty HSE Officer**

Role / Designation: Chief Safety Officer (HSE)

Authorization Record ID: AUTH-20260815-775E09

Authorized At: 2026-08-15 17:27:11 UTC

Approval Notes: Authorized for Substation Sector Emergency Drill.

DEMONSTRATION SIGNATURE BLOCK:

✓ **AUTHORIZED (PROTOTYPE DEMO)**

Duty HSE Officer

Chief Safety Officer (HSE)

DEMO SIGNATURE — NOT A REAL SIGNATURE

(Ready for PKI Digital Signature Integration)